

2017-Ph-H1-Q6

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

A force acts for 2 s and changes a body's velocity. What is the impulse?

 **Key Concept:**

Impulse = change in momentum:

$$J = Ft = \Delta p = mv - mu$$

Given:

Force = 2.0 N, time = 2.0 s

$$J = 2 \times 2 = 4.0 \text{ Ns}$$

 **Final Answer: C — 4.0 N·s**

 **Revision Tips:**

- Impulse = area under force–time graph

- Impulse changes momentum:

$$Ft = \Delta p$$
